Mobile Application Development

(Assignment # 2) (Frontend & Backend Communicating with each other)

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SmartDine — Backend Server

A Node.js + Express REST API backend for the SmartDine restaurant mobile app.

How to Run

1. Install dependencies

Terminal 1: cd smartdine\smartdine\backend

npm install

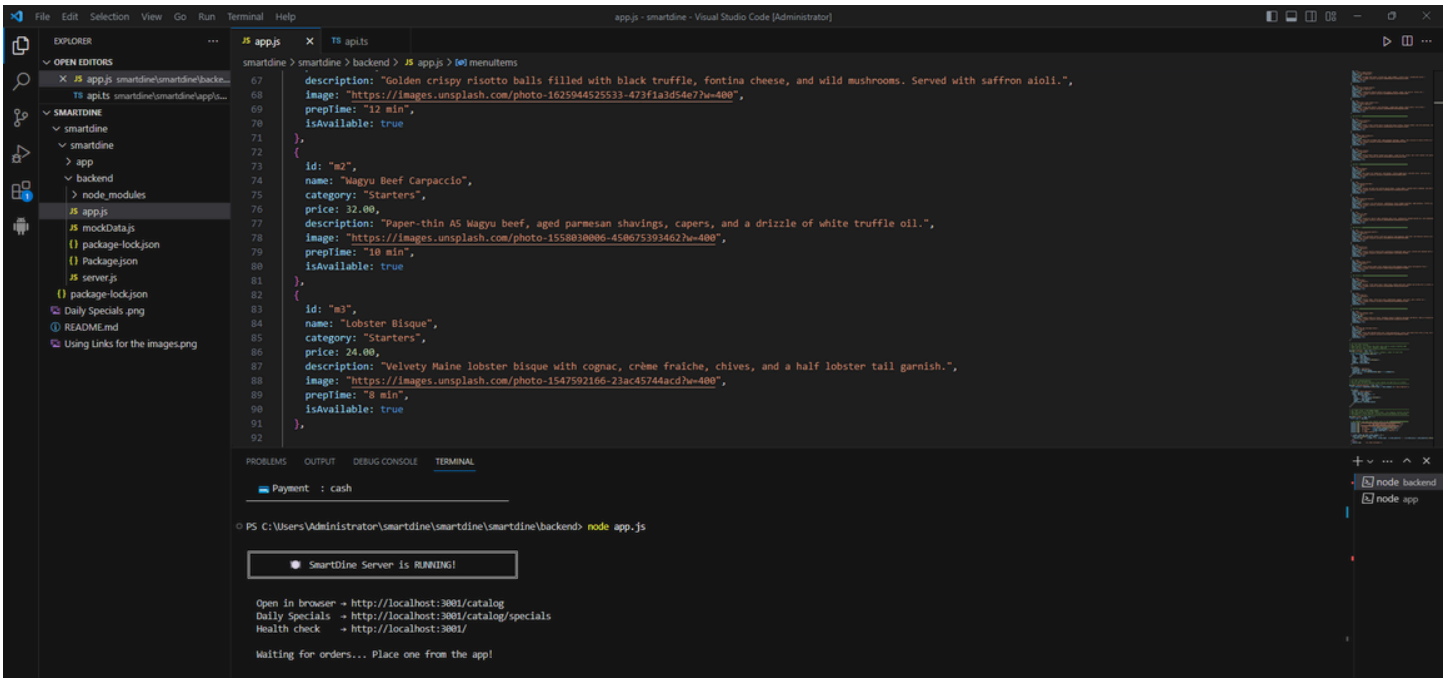
2. Start the server

node app.js

You will see this in your terminal:

```
╔═══════════════════════════════════════════╗
║   🍽️  SmartDine Server is RUNNING!        ║
╚═══════════════════════════════════════════╝
```

Proof of Server Running:



Open in browser → http://localhost:3001/catalog

Daily Specials → http://localhost:3001/catalog/specials

Health check → http://localhost:3001/

FOR FRONTEND :

Open Another Terminal and : cd smartdine\smartdine\app

npm install

npx expo start

Assignment 2 — Required Endpoints

Q1 — Hardcoded Menu Catalog (open these in your browser)

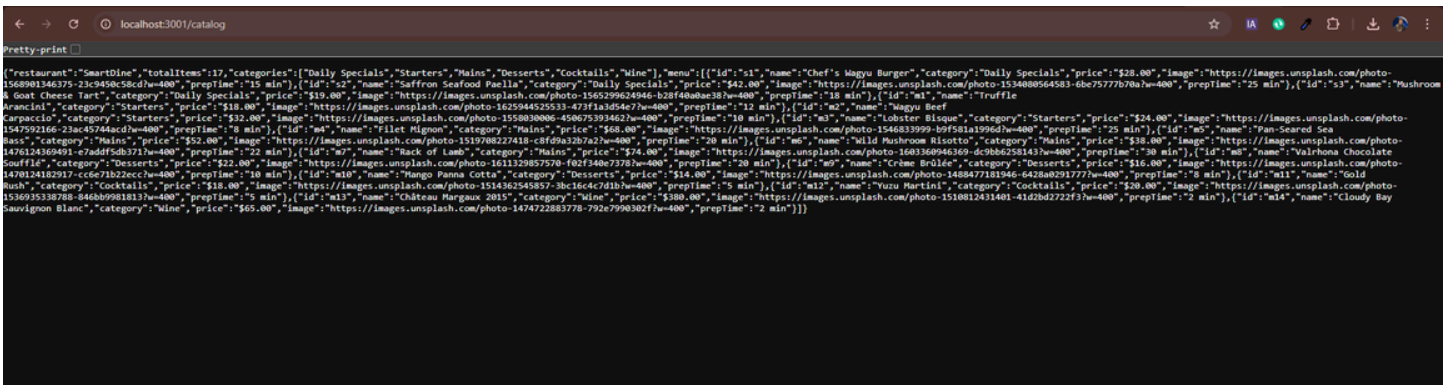
URL	What it shows
http://localhost:3001/catalog	All 14 menu items with name, price, category, image
http://localhost:3001/catalog/specials	Only the 3 Daily Specials items

Q2 — Image URLs

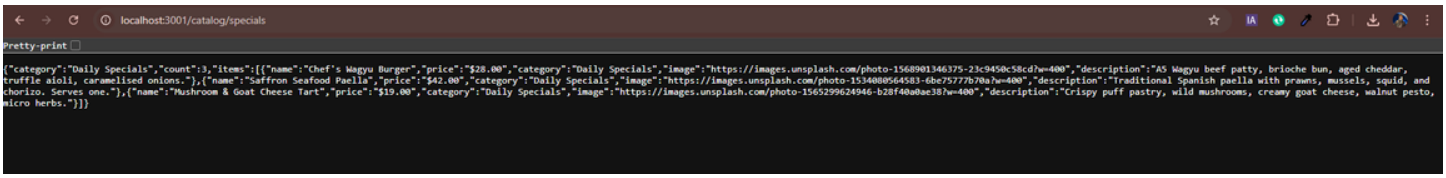
Every item in the catalog has an image field with a direct URL from Unsplash. No local files needed — the app loads images straight from the internet.

Example:


```
{

  "name": "Filet Mignon",

  "category": "Mains",

  "price": "$68.00",

  "image": "https://images.unsplash.com/photo-1546833999-b9f581a1996d?w=400"

}
```

Daily Specials :



Q3 — Order Logger (watch your terminal)

Send a POST request to /order from the React Native app by tapping "Place Order".

The terminal will print:

```
╔═══════════════════════════════════════════╗
‖        🍽️ NEW ORDER RECEIVED — SmartDine        ‖
```

🕐 Time : 2:45:30 PM

👤 Customer : Alex Rivera

🪑 Type : dine-in

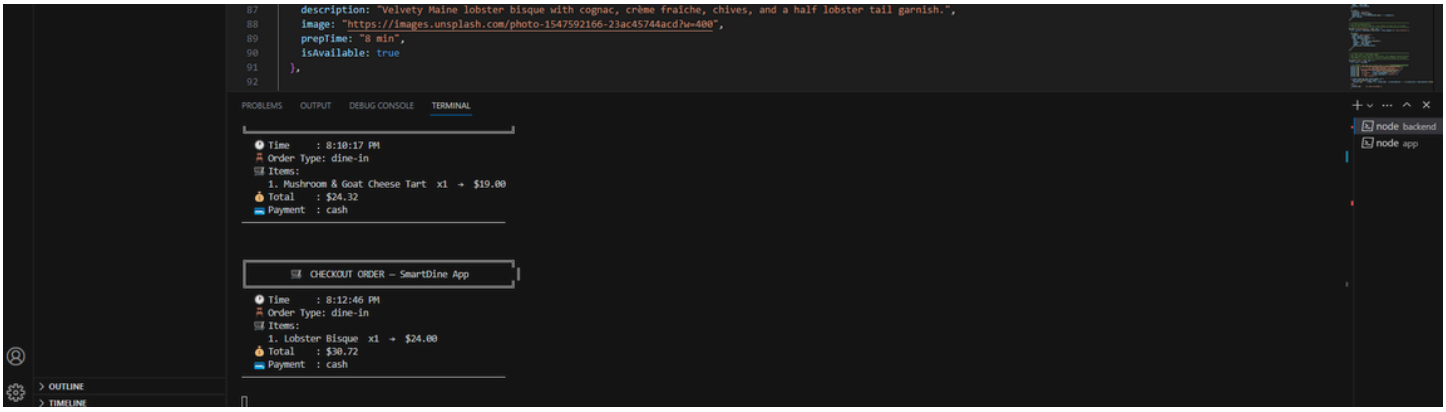
🛒 Items Ordered:

1. Filet Mignon x2 → $136.00
2. Truffle Arancini x1 → $18.00

💰 Total : $154.00

💳 Payment : card

Message on backend after receiving order:



This **proves the frontend and backend are communicating**.

All Available Endpoints

Method	URL	Description
GET	/	Server info + list of all endpoints
GET	/catalog	Full menu with images (Q1 + Q2)
GET	/catalog/specials	Daily Specials only (Q1)
GET	/menu	Menu for the app (supports ?category= and ?search=)
GET	/menu/:id	Single menu item by ID
POST	/order	**Order Logger** — prints to terminal (Q3)
POST	/checkout	Full checkout — also prints to terminal
POST	/login	Login with { email, password }
POST	/register	Register with { name, email, password }
POST	/validate-coupon	Validate a coupon code

Test Credentials

Email: alex@example.com

Password: password123

Test Coupons

Code	Discount
CHEF20	20% off
SAVE10	10% off
BRUNCH24	$15 off

Menu Categories

- **Daily Specials** — 3 items (today's featured dishes)
- **Starters** — 3 items
- **Mains** — 4 items
- **Desserts** — 3 items
- **Cocktails** — 2 items
- **Wine** — 2 items

Project Structure

```
backend/
├── app.js          ← Single server file (all data + all routes)
├── server.js       ← Full server (splits data to mockData.js)
├── mockData.js     ← All seed data for the full server
├── package.json    ← Dependencies
└── README.md       ← This file
```

Tech Stack

- **Node.js** — JavaScript runtime
- **Express.js** — Web framework for handling HTTP routes
- **cors** — Allows the mobile app to connect to this server
- **express.json()** — Parses incoming JSON request bodies automatically